

Adaptive Learning and Correlative Assessment of Differential Usage Patterns for Students with-or-without Learning Disabilities via Learning Analytics

MASOODA M. MODAK, Research Scholar, SP.I.T., Mumbai, Assistant professor, SIES Graduate School of Technology, India
PRACHI GHARPURE, Director, SVKM's NMIMS University, Indore Campus, India
SASIKUMAR M, Director, CDAC,Santacruz, India

Learning Disabilities (LD) can be categorized into logical, analytical, grammatical, vocabulary, sequential, and inference disabilities. Analysis of such disabilities assists students to identify and strengthen their weak areas. A wide variety of analysis models is proposed by researchers to perform such tasks, but most of these models are highly complex and cannot be scaled for multimodal parameter sets. To overcome these issues, this text proposes a model for correlative assessment of differential usage patterns in students with-or-without learning disabilities via multimodal analysis. The proposed model initially collects real-time inference sets for students with Learning Disabilities (LD) and without LDs. These sets consist of question-specific recorded responses for "Addition," "Carry Propagation," "Basic to Advanced Grammar," "Direct, Inference and Vocabulary Comprehension," "Finding odd-man-out," "Sequencing," and "Pseudo and Sight Spelling" for different question sets. Answers to these questions and their metadata were processed via a correlative engine that assisted in evaluation of correctness, time needed per question per category, number of skips, number of revisits, and unanswered ratio for different students. This evaluation was combined with temporal analysis to identify per-category progress of students. Based on this progress, students were either upgraded to next level or given lower-level questions, which assisted them to incrementally improve their grades. The model proved that the performance of LD students is 55% less than the non-LD students and an average of 18 LD students have achieved an average of 33% of improvement after having multiple attempts of the adaptive lessons. The model uses a correlation function, which enables to identify answering patterns of LD and non-LD students with 98.4% accuracy, thus can be used for clinical scenarios.

CCS Concepts: • **Computing methodologies** → **Machine learning**; • **Information systems** → *Web data description languages;*

Additional Key Words and Phrases: Learning disability, correlation analysis, multiple attempt analysis, multimodal analysis

All authors contributed equally to this research.

Authors' addresses: M. M. Modak (Corresponding author), Research Scholar, SP.I.T., Mumbai, Assistant professor, SIES Graduate School of Technology, Nerul, Navi Mumbai, Maharashtra, India, 400706; e-mail: masoodamodak29@gmail.com; P. Gharpure, Director, SVKM's NMIMS University, Indore Campus, Indore, Maharashtra, India; e-mail: prachi.gharpure@gmail.com; M Sasikumar, Director, CDAC, Santacruz, Mumbai, Maharashtra, India; e-mail: the.little.sasi@gmail.com.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
© 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

2375-4699/2023/12-ART258 $15.00
https://doi.org/10.1145/3632365

ACM Reference format:
Masooda M. Modak, Prachi Gharpure, and SasiKumar M. 2023. Adaptive Learning and Correlative Assessment of Differential Usage Patterns for Students with-or-without Learning Disabilities via Learning Analytics. *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 22, 12, Article 258 (December 2023), 25 pages.
https://doi.org/10.1145/3632365

1 INTRODUCTION

One of the most prevalent cognitive barriers to communicating verbally or in writing is a **disorder of specific learning (SLD)**. It also manifests itself in problems with reading, writing, and listening. In India, the occurrence of learning disabilities varies from 2.16% to 30.77% [1]. Specific learning disorders might take the form of dyslexia, dysgraphia, or dyscalculia. Dyslexia is a disorder that makes schoolwork challenging. Since dyslexia is a neurological disorder, its effects will vary, depending on the individual who suffers from it. It has lifelong consequences on a child's ability to express themselves verbally and in writing. Due to confusion between letters and words, they read slowly and incorrectly. The experts are of the opinion that, even when the same type of disability is involved, there is no one solution that can meet all individual needs; as a result, using external tools tailored to particular special needs is preferable to create embedded facilities; and that special attention should be paid to accessibility of the content uploaded to the LMS [2]. Group learning activities must be made accessible and modified for students with and without difficulties who are participating in e-learning [3]. There are specific issues with the current e-learning online models, gaming apps that still exist, such as personalizing content based on the needs of the students and identifying students who have learning disabilities [4, 5]. Students taking online lessons while wearing an EEG headband will provide real-time e-learning data. The CNN model proposed in Reference [5] will divide the students into grades and aid in the creation of an automated framework for the monitoring of a student learning curve, offering suggestions for improving the course materials for online learning. These models provide a very high level of accuracy for real-time applications, but they impose additional implementation expenditures. Moreover, these models cannot be used in normal learning settings, because they are invasive; as a result, they are inappropriate for usage in general e-learning environment.

Children who struggle with dysgraphia sometimes produce slanted, distorted, or incorrectly capitalized writing. Children who struggled with spelling tended to write or copy slowly. Children with dysgraphia often struggle with number sense and seeing patterns in mathematical formulas and situations. Dyscalculia makes it hard for children to accomplish even basic arithmetic, since they cannot grasp how operators and numbers work. Thus, each child with a different learning difficulty has a different type of struggle. This implies that it is essential to capture the learning characteristics of each of these types of learners. But the machine learning models employed in previous works are black box models, making them unable to determine the features on which the presence or absence of a learning disability is determined.

For the purpose of LD detection, several research studies employ unofficial techniques. The methods employed in these research studies to identify **learning disabilities (LD)** in children rely on the provided checklists, which contain the symptoms and signs of the specific LD. Many times, a certain symptom listed on the checklist is applicable to a child, but he is unaware of it. The students often struggle to admit that they experience one or more of the symptoms listed in the checklist. Furthermore, the list of signs and symptoms is static and could leave out a lot of different indications of the various learning difficulties. Given the above-mentioned difficulties, we require a system that can recognize LD and understand the learner's learning process. The system must be capable of analyzing LD symptoms, determining how they relate to one another, and diagnosing the presence of a learning disability.

Thus, the motivation of this research is to provide learning according to the dynamic requirements of the student. A unique detection model is employed to detect the presence or absence of learning disability based on the dynamic learning profile of the student. Once the difficulty is detected, the student will learn according to his capabilities and preferences using the learning analytics engine that uses the proposed statistical model for deciding the adaptive content. The adaptive content presented to user students with disabilities makes use of e-learning that has LMS software packages. The proposed model provides learning settings that are helpful to the success of children with learning disabilities. These settings recognize and accommodate the unique needs of each student. The proposed system has an analytical engine that analyzes the continuous performance of the dynamic learning pattern of the student. For analytical analysis, classifiers such as Decision Tree, SVM, Logistic Regression, and KNN are employed. The analytics engine enables the LD-detected student to study according to his learning preferences, thereby inspiring them to overcome their obstacles to learning. The article also presents the correlative assessment of the usage patterns of students with and without learning disabilities in terms of parameters such as the number of correctly answered questions, amount of time taken, the number of skips, the number of revisits, percentage of improvement or no improvements in comparison of LD versus non-LD. Based on this observation, Section 3 discusses methodology of research. Section 4 discusses the design of a model for multimodal assessment of differential usage patterns among students with and without learning disabilities. Section 5 discusses adaptive learning through dynamic profiling using statistical modeling, Section 6 presents model validation with accuracy, precision, and recall of LD detection. Section 7 presents the analysis of the reattempts for the comparison of revisits, skips, as well as the typical time delay required by LD students to improve their performance. When the proposed model's performance was compared to that of several state-of-the-art models, it was shown to perform better; this facilitates its use in clinical settings. This article concludes with some insightful observations regarding the proposed model and offers a number of suggestions for enhancing it.

2 LITERATURE REVIEW

Various works are been proposed and implemented in this area. The studies in References [6, 7] offer a comprehensive overview of how WebVR might be applied to dyslexia instruction. The study also discusses how spatial data may be used as an input. For richer and more dynamic display, relationships among data can be rendered into a 3D environment, which refers to a variety of connections, interactions, or relationships between distinct elements that can be represented or visualized in a **three-dimensional (3D)** virtual space. Datasets used in machine learning frequently contain a number of characteristics or properties that can have an impact on the result or target variable. The performance and predicted accuracy of the model may be impacted by the complex interactions between these features. However, standard two-dimensional plots might be difficult to visualize high-dimensional data (data with numerous features). We may be able to obtain a better understanding of how various aspects interact with one another by simulating these interactions in a 3D environment. The work suffers from absence of efforts or elements that would establish connections or interrelationships among the relationships that have been identified or captured. This interrelationship could help to diagnose the presence of a learning disability.

Eye-tracking motions have been demonstrated to be a reliable indicator of reading difficulties in children. Dyslexia can be identified with the use of eye-gaze behavior analysis. The study in Reference [8] aims to show the capability of gaze-based predictions, where the predictive analytics is used to improve user success and enhance user experience. This study intends to promote the development of adaptive learning according to the identified learning difficulty. According to the relevant results in Reference [9], a significant improvement has been illustrated by applying

direct human eye-gaze indication, which quantitatively points out that using eye gaze can be more efficient than other human-action based inference approaches.

The design of the detection system used in Reference [10] employs **Logistic Regression (LR)** and Deep Forest models. The inclusive customized learning environment has a prototype app for use by people with special needs. Predicting the likelihood of a child to be a having learning disability is the major goal of this approach. Using the disability model, a profile is generated for a student who has a disability. The disability model enables the best virtual learning object to be suggested based on the learner's preferences and actions.

For students with learning difficulties, **intelligent assistive technologies (IAT)** are created to offer individualized help and accommodations to improve their academic progress. These tools make use of different artificial intelligence techniques, machine learning techniques, and adaptive algorithms to meet the unique demands of students who face varied learning difficulties. Assistive technology to help those with visual and learning disabilities is also used in research work proposed by researchers in Reference [11]. Concatenation synthesis is used in this suggested assistive device's **Text To Speech (TTS)** conversion and **Optical Character Recognition (OCR)** processes. The main conclusion was that instructors should use technology to teach reading to boost LD children's reading abilities. Participants in the experimental group consistently outperformed their contemporaries. According to research, the most efficient technology for teaching one reading skill utilized gaming or drill-and-practice formats.

Speech fluency is one of many language development processes that might be impacted by learning disabilities. The majority of currently used disfluency detection methods heavily rely on expensive and infrequently available human-annotated data. The authors in Reference [12] investigated how to use active learning into the fine-tuning process to close the performance gap. With a limited annotated dataset, active learning may reduce the drawback of self-supervised learning by focusing on the most significant instances to label by the fusion of self-supervised learning with active learning.

The work in Reference [13] proposes models that utilize **Augmented Reality (AR)**, social and collaborative technologies, and the modification of learning environments. But these models are not able to assist a large variety of students because of their infrastructure and cost requirements. These models also require students to learn about technologies before using them, which further restricts their scalability. Similar models are proposed in References [14, 15], wherein, intervention-based approaches, such as **individualized reading assistance (IRA)**, and the utilization of cost-effective technologies are employed in order to make performance enhancement models more accessible and scalable. The drawback of this work is that the same intervention techniques are used in a variety of contexts. This technique does not consider the particular situations and requirements of various children or groups.

Extensions to these models are discussed in References [16, 17], wherein methods such as special attention for LD students, inculcation of self-determination among students, and **intervention-based models (IM)** for continuous assistance are proposed. Out of these models, the intervention model is highly efficient, because it showcases high student throughput performance when applied in real-time scenarios. But these models are only aimed at improving the reading fluency related to dyslexia.

Both the students' learning difficulties and criteria for measuring their performance, such as memory, attention, and language, are provided by the authors in References [18, 19]. The strengths and weaknesses of the learner, as well as potential roadblocks to e-learning, may be identified in this way. Researchers will keep tweaking the prototype and adapting it to new impairments in an effort to boost students' performance in a digital classroom [20, 21]. This web software was made to serve as a quick diagnostic tool for learning disabilities. Each of their control and experimental

groups has an equal number of youngsters. The children in both sets of tests will have 10 activities to complete that will measure their literacy, numeracy, and mental acuity. Results showed that children with learning disabilities gave wrong answers at a higher rate than the control group. The number of correct responses and the amount of time spent on each assignment will be used to make these selections [22, 23]. Children who may have learning disabilities may be identified using ICT if an ontology-based ICT tool is used. It is very beneficial for both parents and educators, since this paradigm involves testing for learning impairments in youngsters. Furthermore, it provides appropriate learning opportunities [24, 25] for different use cases. The drawback of the above approaches using web software is that they lack interaction with the child being examined, which is a crucial aspect of diagnosing learning difficulties. The web-based program in these studies is not able to imitate how trained professionals can monitor nonverbal cues, clarify questions, and create a more comfortable and suitable setting for assessment.

Researchers have created a broad range of analytical models to carry out SLD identification tasks. According to the findings by the authors in Reference [26], who compared two groups—students with learning disabilities and those with no learning disabilities—students with LD reported worse social and academic personalities. Additionally, they reported lesser levels of motivation and effort put into their academic work, as well as perceiving their mood as more negative. Emotional behavior comparisons to the non-LD peer group did not reveal any noticeable changes at the same time. The work in Reference [27] makes a thorough argument from a variety of theoretical angles that five factors, including learning commitment (the learner's commitment to participating in and excelling at the learning process), course objectives (how closely the learner's expectations fit the objectives of the curriculum), learning actions (the precise behaviors, attitudes, and tactics that students use to understand and retain knowledge), learning soul (the learner's determination, passion, and sense of connection to the subject), and learning external environment, will affect the degree of learning disabilities and learning adaptability among college students. The research concluded that for better learning, adaptation of content to learner characteristics is important and needed. Research in Reference [28] suggests a model that is built using cellular automata and association rules in conjunction with fuzzy logic and machine learning. The benefit of the suggested approach is that it aids in identifying these students' needs, preventing commonly used procedures from influencing their approach to learning. Therefore, it is important to recognize that every child learns at a different speed. With **Hybrid Neuro-Fuzzy (HNF)** methods, however, the vast majority of these models are very sophisticated and cannot be scaled for multimodal parameter combinations.

In References [29, 30], student feedback and student misbehavior analyses are used to estimate the levels of LD and non-LD that are present in certain students, further demonstrating the effectiveness of such models. Finding out if a student has a learning difficulty is the aim of this research. The level of accuracy of these models is somewhat constrained by the use of data collection and analysis techniques, each of which has certain limitations. According to the studies described in References [31–33], continuous supervision, **1-Dimensional Convolutional Neural Network (1D-CNN)** models based on electroculography, and app-based scanning of students may all be used to increase classification accuracy in real-time settings. The development of these improvements is directed by the intention to contribute to improving the system's overall performance. Although these models are quite precise, scaling them up would require the installation of more components, which is not feasible. Because of this, they are unable to be used in real-world applications. A gamification model was suggested in studies that were published in References [34, 35] with the aim of enhancing scalability as well as classification performance in real-time circumstances. In this model, **k-Nearest Neighbours (kNN), Support Vector Machines (SVM)**, and **Random Forest (RF)** are all combined. This action is being conducted in an effort to overcome the limitation that was previously specified in the paragraph. The various extensions that may be

made to these models are discussed in Reference [36], in which the **Test of Mathematical Abilities (TOMA)** or the Test of Symbolic and Nonsymbolic Numerical Magnitude Processing, both of which were carried out, assisted in improving the context-specific performance of these models under clinical conditions. This refers to characteristics that improve the performance of machine learning models when they are used in practical clinical contexts.

The majority of these models, however, rely on machine learning approaches, which enhances their applicability in scenarios where the context significantly influences the outcome. Additionally, the characteristics that indicate learning disabilities are concealed inside a black box. Many students may be using e-learning platforms without the required learning abilities. For those who struggle with learning, there are several online resources. However, they fail to take into account and adapt to the unique learning properties. Each student with a learning impairment may have a unique learning disability. For instance, one dyslexic learner could fail to read written text, while another may struggle with letter and number reversals. Many times, the student is unaware that he is facing a particular LD symptom. Sometimes students are reluctant to acknowledge that they experience one or more LD symptoms. To address these concerns, this research then provides a paradigm for the detection of learning disabilities using dynamic learning characteristics using a detection model. Further, this model will also provide statistical modeling of the learning behavior correlative evaluation of distinctive use patterns in students with and without learning problems based on multimodal analysis. Identifying the different traits that are specific to a certain type of learning disorder is important to recognize a learning disability. Determining the features of learning disabilities according to age or grade is the first step towards identification of learning disability. The student will be able to learn in accordance with his or her learning requirements and preferences once the disability has been identified, which might result in improved performance for the learner with LD.

To validate its efficacy, the suggested model's performance was also assessed in the context of several real-time scenarios and compared with existing approaches. The research approach to accomplishing the objectives is covered in Section 4, which presents the design of the proposed correlative assessment of differential usage patterns for students with or without learning disabilities via multimodal analysis. Section 5 validates and hypertunes the model. Section 6 compares the proposed model's performance to that of current approaches in terms of precision, recall, accuracy, and latency metrics across a variety of real-world use cases. The last section of this research offers some concluding thoughts on the suggested paradigm and how it may be improved for therapeutic applications.

3 METHODOLOGY OF RESEARCH

The goal of the research is to create a method for identifying a learning disabled person's learning profile. A behavioral analysis engine is created to examine how people with learning disabilities learn. The engine will make an effort to determine the learner's preferences to try to record the learner's behavior when they are learning. All of this student-related data will be included into a learner model that will be created. Throughout the learner's learning process, the analytics engine will dynamically update the learner model. When given the characteristics and features from the learner's learning pattern, the model would be trained using machine learning to identify the learning disability.

(1) Creation of a learner model based on the dynamic properties of learning. The students are categorized as LD/NonLD using the machine learning model. Example: Consider a student A, who begins exploiting the adaptive learning platform. On the basis of her quick comprehension and high engagement with the topic, the model first labels her as Non-LD.

(2) The Behavioral Analysis Engine will update this model to capture the learner's evolving learning features. The student goes through this process as they are learning. Example: The engine observes the student A's learning rate has slowed and her engagement level has fallen as she moves through her curriculum. It makes the appropriate adjustments to her learner model.
(3) This engine will show the learner the adaptive content, assess how well they did, and update the model if necessary. The modified learner profile will be reflected in the learner model. Example: A's slower learning rate is recognized by the platform, which then provides her with additional practice activities on difficult subjects. The material adjusts to fit her new pace once she has shown improvement.
(4) The performance evaluator will assess the performance and perform correlative assessment in accordance with the different performance metrics discussed in the text based on the answers to the questions. Example: The system keeps track of A's quiz results, the amount of time she spends on assignments, and how much she interacts with the supplemental resources to determine how well she did. It shows a strong positive correlation between her use of extra resources and her improvement in quiz results.
(5) The model is validated for robustness and improvements over time. The model is refined continuously for new data and knowledge.

4 DESIGN OF THE PROPOSED CORRELATIVE ASSESSMENT OF DIFFERENTIAL USAGE PATTERNS FOR STUDENTS WITH-OR-WITHOUT LEARNING DISABILITIES VIA MULTIMODAL ANALYSIS

Based on the review of existing LD identification techniques, it can be observed that a wide variety of analysis models is proposed by researchers to perform this task, but most of these models use static parameters and are not able to reflect the dynamic learning behavior of a student. These techniques are highly complex and cannot be scaled for multimodal parameter sets. This section discusses the design of a model for multimodal assessment of differential usage patterns among students with and without learning disabilities to address these issues. Figure 1 depicts the model's flow, where it can be seen that the proposed model initially collects real-time inference sets for students with and without LD. These sets include question-specific recorded responses for "Addition," "Carry Propagation," "Basic to Advanced Grammar," "Direct, Inference and Vocabulary Comprehension," "Finding odd-man-out," "Sequencing," and "Pseudo and Sight Spelling" for a variety of question sets. Answers to these questions and their metadata were processed by a correlative engine that aided in evaluating correctness, time required per question per category, number of skips, number of revisits, and unanswered ratio for various students. This evaluation was combined with a temporal analysis to determine the students' progress by category. Based on this progress, students were either promoted to the next level or given lower-level questions to help them improve their grades incrementally for real-time scenarios. To perform these tasks, the model extracts a series of analytical features from collected data samples. These features include:

— Correctness (C), which is estimated via Equation (1) and assists in finding total questions that were correctly answered for individual categories.

$$C = \frac{1}{N_q} \sum_{i=1}^{N_q} C_{a_i} \ldots, \tag{1}$$

where N_q represents number of questions in the given category, while C_a is a flag, which is set to 1, if student has answered the question correctly, else it is 0 for incorrect responses.

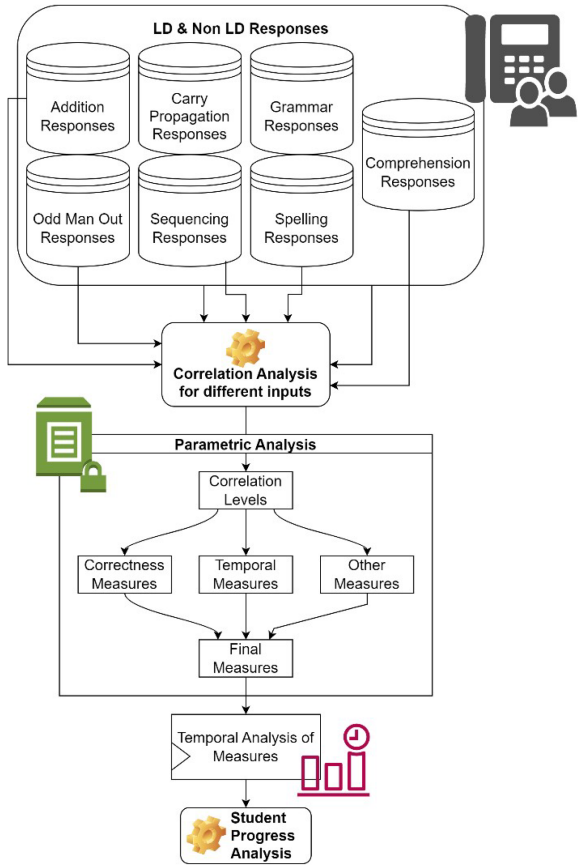


Fig. 1. Design of the proposed temporal model for correlative analysis.

— Time needed per question (T), which is estimated via Equation (2) and is a measure of speed-of-answering different question types.

$$T = \frac{1}{N_{cq}} \sum_{i=1}^{N_{cq}} TS_{\text{complete }_i} - TS_{\text{start }_i} \ldots, \tag{2}$$

where $TS_{complete}$ and TS_{start} represent the timestamps for completing and starting the question attempts, whileN_cq represents number of correctly answered questions.

— Number of skips(N_s) is evaluated via Equation (3) and assists in identification of skipping probability of students for different question types.

$$N_s = \frac{1}{N_q} \sum_{i=1}^{N_q} S_{a_i} \ldots, \tag{3}$$

where S_{a_i} represents total number of questions skipped by the student for individual categories.

— Number of revisits (N_r) is estimated via Equation (4) and assists in identification of question types for which students are taking multiple chances.

$$N_r = \frac{1}{N_q} \sum_{i=1}^{N_q} R_{a_i} \ldots, \tag{4}$$

where R_{a_i} are total number of revisits for a given question type.

— Unanswered ratio for different students (U_a), is evaluated via Equation (5) and assists in identification of percentage of questions unanswered by students for individual category types.

$$U_a = \frac{1}{N_q} \sum_{i=1}^{N_q} U_{a_i} \ldots, \tag{5}$$

where U_a is a flag, which is set to 1, if student has not-answered the question, else it is 0 for answered questions. These features are combined to form a per-category vector via Equation (6).

$$F_{\text{cat}} = C \bigcup T \bigcup N_s \bigcup N_t \bigcup U_a \ldots \tag{6}$$

These features are collected for LD and Non-LD students and fed as input to the detection model. The detection model is an ensemble of classifiers (KNN, Random forest, SVM, Linear Regression). The data is collected from LD-certified students and non-LD students from grades 6, 7, and 8.

5 ADAPTIVE LEARNING THROUGH DYNAMIC PROFILING USING STATISTICAL MODELING AND LEARNING ANALYTICS

For the student to acquire knowledge in accordance with his or her changing learning characteristics, adaptive learning is crucial. Using questions from the **Question Bank Generator (QBG)** layer, the system used to identify learning disabilities includes a layer that estimates student-level behavior. Each question receives a separate evaluation and update for this level. Multiple tests are given to both LD and non-LD children, and their responses are recorded. These responses are saved for each subject type, and Equation (7) is used to get an average differential score for each subject.

$$ADS_{Student}\,(Sub) = \frac{\sqrt{\left(\sum_{i=1}^{N_{eval}} \left[C_{score_i} - \sum_{j=1}^{N_{eval}} C_{score_j}\right]^2\right)}}{N_{eval}} \ldots, \tag{7}$$

where C_{score} represents correctness score for the given question type and is evaluated via Equation (8) as follows:

$$C_{score_i} = \frac{N_{QC_i}}{N_{QC_i} + N_{QI_i} + N_{NA_i}} \ldots \tag{8}$$

Based on this score for LD and non-LD students, an incremental score is evaluated for each subject via Equation (9) as follows:

$$I_{score}\,(Sub)_i = \frac{ADS_{NLD}\,(Sub) - ADS_{LD}\,(Sub)_i}{ADS_{NLD}\,(Sub)} \ldots, \tag{9}$$

where ADS_{LD} & ADS_{NLD} represent average differential score for LD and non-LD students and is evaluated at student level (i) for each LD student. This assists in estimation of average performance

deviation for LD students when compared with their normal counterparts. To evaluate number of questions per subject, a probability score at subject-level is initialized via Equation (10).

$$P_s\,(Sub) = \frac{1}{N_{sub}} \ldots, \tag{10}$$

where N_{sub} represents number of subjects for which this model is being deployed, and P_s represents probability score for that subject. This score is modified via intensity and polarity of incremental score. To perform this modification, the following conditions are evaluated:

— If $I_{score} \leqslant 0$, then this indicates that non-LD students are performing at-par with LD students for the given subject and do not need any additional evaluation, thus, subject probability score is modified via Equation (11).

$$New\,(P_s\,(Sub)) = P_s\,(Sub) - \frac{|I_{score}|}{N_{sub} * Max\left(\bigcup_{i=1}^{N_{sub}} I_{score}\right)} \ldots \tag{11}$$

— If $I_{score} > 0$, then it indicates that student needs to improve their performance for the given subject, thus subject-level probability score is modified via Equation (12) as follows:

$$New\,(P_s\,(Sub)) = P_s\,(Sub) + \frac{2 * |I_{score}|}{N_{sub} * Max\left(\bigcup_{i=1}^{N_{sub}} I_{score}\right)} \ldots \tag{12}$$

Equation (13) is used to calculate the number of questions for each subject based on the new subject-level probability.

$$N_q\,(Sub) = \frac{P_s\,(Sub)}{\sum_{i=1}^{N_{sub}} P_s\,(i)} * T_q \ldots, \tag{13}$$

where T_q represents total number of questions to be attempted by the student and are decided by the system designer.

Based on this process, question sets are generated, their responses are recorded, and, using these responses, behavior of students is evaluated for dynamic question bank generation and control. Once a student has completed an iteration of examination, then the system recommends improvement areas. These improvement recommendations are used for continuous performance enhancement of the student for a particular subject. At the end of this evaluation, a **non-LD metric (NLM)** or percentage improvement is estimated via Equation (14).

$$NLM = \frac{Cat(NonLD)}{TC} \ldots, \tag{14}$$

where Cat(Non LD) represents the number of categories for which the student is classified under non-LD category, and TC represents total number of evaluation categories. Students are upgraded to next level for non-LD categories, if NLM>0.5, which indicates that student has outperformed in at least 50% of question categories. Based on this above complete process, students are categorized into LD and non-LD types, and their learning performance is continuously improved for real-time scenarios.

Once the child has attempted the questionnaire for various subjects, the detection model will detect in which subjects the child is facing a learning difficulty. The statistical modeling will be used to decide the level of difficulty the child is facing in each subject using the $I_{scorevalue}$ and thereby the amount of improvement needed. Once the child has attempted the lessons multiple times, the learning analytics engine will update the learner model to store the updated learner profile of that child. The learning analytics engine enables to predict the status of the child on his first attempt. If the child is detected as suffering from a learning disability, then the LA engine also

recommends the number of re-attempts needed to have a substantial amount of improvement in that child for a particular lesson.

These features are collected for LD and non-LD students and stored to perform a correlation process. Similar features are extracted for new students, and then a statistical measure such as correlation is estimated with LD and non-LD features via Equation (15).

$$\text{Corr} = \frac{\sum_{i=1}^{N_f} \left(\begin{array}{c} F_i - \\ \text{avg}(F) \end{array} \right) \left(\begin{array}{c} F(\text{ New })_i \\ -\text{avg}(F(\text{ New })) \end{array} \right)}{\sqrt{\sum_{i=1}^{N_f} \left(\begin{array}{c} F_i - \\ \text{avg}(F) \end{array} \right)^2}} \ldots \tag{15}$$

where F and $F(New)$ represent features of already stored samples and new student samples. This correlation is estimated for LD and non-LD features under individual question categories. Based on this correlation level, students are having a high chance of falling into category-specific LD and non-LD students via Equation (16).

$$Student \in LD, if \;\; Corr(LD) > Corr(NonLD) \;\; else \;\; Student \in NonLD \ldots \tag{16}$$

If the features of new student is highly correlated with features associated with learning disabilities for a specific question type, then the analytical engine runs the detection algorithm to detect the presence/absence of learning difficulty in each subject so accordingly, the difficulty level for that question type is reduced, and students are re-analyzed via the same process.

To validate this performance, the model was evaluated in terms of accuracy of mapping, precision levels, recall levels, and delay needed for mapping under real-time scenarios. This evaluation was done for different use cases in the next section of this text.

6 MODEL VALIDATION

Within the framework of our binary classification task, we evaluate the **Random Forest (RF)** classifier's performance in this section. We use a variety of indicators to conduct a thorough evaluation with the goal of determining how well the classifier can distinguish between the two classes.

To allow LD and non-LD students to participate in the various lessons, the system was made available online. The lessons were made available to the 6th-, 7th-, and 8th-grade LD or non-LD students. The LD students who attempted the classes were identified as having a learning impairment and were connected to special schools or remediation facilities. Of the 220 students that attempted the classes, 137 were non-LD and 83 were LD. This made sure that accurate responses from LD students were captured to create a strong training and testing dataset and, ultimately, a model for more accurately identifying LD in a given subject. The dataset's correctness and reliability were ensured by the students' real responses, including those who had learning problems and those who did not.

A manual dataset was created by collecting the responses of 220 students who participated actively in 13 subject areas. Then, their performance in these subjects was assessed using this dataset. The remediation center was contacted to approach the LD children, since they had received official certification of their condition. The dynamic learning profile was extracted from their responses, and the ensemble algorithm was used to classify them into LD and non-LD groups. To determine the final classification performance, the classified sets were compared with ground-truth information. A total of 2,500 samples were used for classification, and they were split up into a group of 70:30 samples. We employed an RF classifier to classify the children with or without learning disability. Based on this arrangement, 1,750 samples were utilized for training, while the remaining 750 samples were used for testing and validation. Then, using the results for **accuracy (A),**

Table 1. Performance Metric for Random Forest Classifier

Metric	Value
Accuracy	73.23
Precision	67.14
Recall	69.43
F1-Score	68.54

precision (P), recall (R), and **area under the curve (AUC)**, all 2,500 samples were used for the parametric analysis. The total number of correct predictions a model makes may be counted using an evaluation metric called accuracy. In the context of binary classification, a form of machine learning problem where the objective is to categorize instances into one of two classes (positive or negative), words such as TP, FP, TN, and FN are frequently employed. The term **true positives (TP)** refers to situations where the model properly predicted the positive class. **False Positives (FP)** are situations where the model predicts the positive class while the true class is really negative. **True Negatives (TN)** are situations in which the model accurately predicted the negative class. **False Negatives (FN)** are situations where the model predicts the wrong class, usually a negative one, when the correct class is a positive one. So, despite the model's negative prediction, the actual class was positive. Accuracy, precision, and recall are some of the evaluation metrics for classification models that are frequently calculated using these concepts. By taking into account several features of a classification model's predictions, these metrics assist in evaluating its performance. In Equation (17), the accuracy calculation is displayed.

$$Accuracy = \frac{TruePositive + TrueNegative}{TruePositive + TrueNegative + FalsePositive + FalseNegative} \ldots \quad (17)$$

Precision is the response to the question, "How often was a model's positive prediction accurate?" Precision is the percentage of outcomes that are appropriate. The precision formula is shown in Equation (18).

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive} \ldots \quad (18)$$

Recall quantifies the proportion of true positives that a model correctly identified. In Equation (19), the recall formula is displayed.

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative} \ldots \quad (19)$$

The accuracy of a model on a dataset is measured by the F-score, also known as the f1-score. It is employed to evaluate binary categorization schemes that label instances as "positive" or "negative." The f1-score formula is shown in Equation (20).

$$f1_{score} = 2 * \frac{Precision * Recall}{Precision + Recall} \ldots \quad (20)$$

The results of our RF classifier evaluation are summarized in the following table:

Table 1 presents the performance of the random forest classifier. To boost the model's performance, we implemented a cross-validation technique called RandomizedSearchCV. RandomizedSearchCV is a popular techniques for hyperparameter tuning in machine learning. It is used to systematically search for the best combination of hyperparameters for a machine learning model. With respect to detection of learning disability and the parameters accuracy, recall, and precision, our classifier should have a better recall, because in case of detecting LD, we should have low

Table 2. Performance Comparison of Random Forest with RandomizedSearchCV

Metric	RF	RF with RandomizedSearchCV
Accuracy	73.23	95.64
Precision	67.14	94.41
Recall	69.43	93.24
F1-Score	68.54	94.42

Table 3. Performance Parameters of the Proposed Model with Other Models

Metric	**RTL [8]**	**LR [9]**	**HNF [26]**	**Proposed**
Accuracy	80.35	74.11	86.38	95.64
Precision	88.03	85.57	90.27	94.41
Recall	86.94	84.5	89.15	93.24
Delay	170.81	120.24	103.44	76.41

false negative rate. Thus, we hypertune the parameter with cross-validations to improve the accuracy, precision, and recall score. The hypertuning was done for the parameters "max_features" and "max_depth." This technique allowed us to efficiently explore a wide range of hyperparameter combinations, resulting in improved model performance metrics.

From Table 2, it can be seen that the performance of random forest classifier has improved by using the hypertuned parameters with the random forest classification. Thus, using the improvement metric, the RF with hypertuned parameters is approximately:

— 14.4% better in accuracy
— 22.08% better in precision
— 12.05% better in recall,
— 20.51% better in F1-score.

7 RESULTS ANALYSIS AND COMPARISON

With the hypertuned Random forest model, a total of 2,500 evaluations were performed, out of which 70% were used for training the model, while 30% each were used for testing and validation purposes. Based on this strategy, the model was evaluated in terms of accuracy of detection, precision, recall, and delay compared with RTL [8], LR [9], and HNF [26]. Table 3 presents a comparison of these parameters for all the above-mentioned models with the proposed model.

From Table 3 and Figure 2, it can be observed that the proposed model is able to improve the accuracy of detection by 14.5% when compared with RTL [8], 19.4% when compared with LR [9], and 9.5% when compared with HNF [26] under different use cases. The accuracy is improved due to the use of high-performance evaluation metrics, which assist in improving LD and non-LD identification performance in real-time scenarios. The proposed model is able to improve the precision of detection by 5.9% when compared with RTL [8], 8.3% when compared with LR [9], and 3.5% when compared with HNF [26] under different use cases. This precision is improved due to the use of correlative mappings, which assist in improving LD and non-LD identification performance in real-time scenarios. Similarly, the proposed model is able to improve recall of identification by 5.5% when compared with RTL [8], 6.4% when compared with LR [9], and 2.9% when compared with HNF [26] under different use cases. The recall is improved due to use of high-performance temporal correlative mapping, which assists in improving LD and non-LD identification performance in real-time scenarios. Similarly, the proposed model is able to improve speed of detection by 19.4%

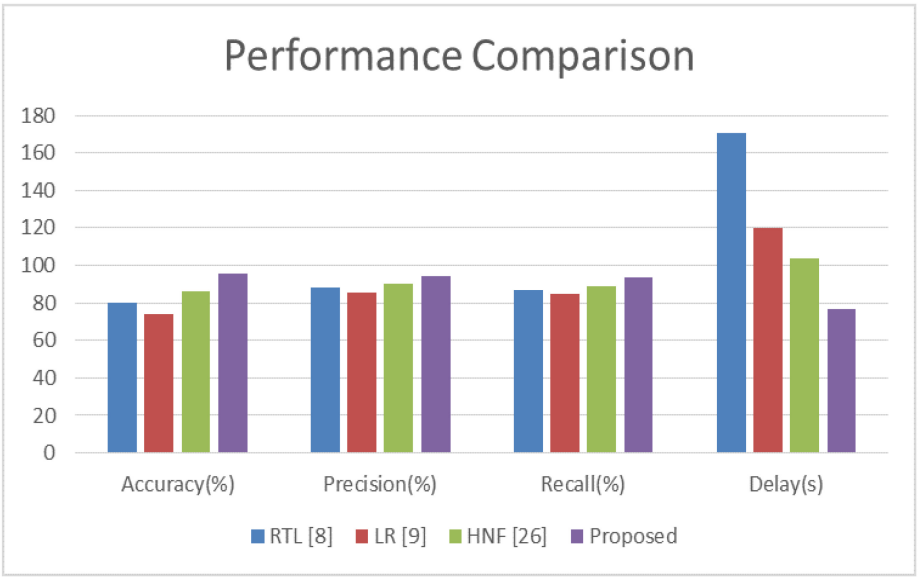


Fig. 2. Performance of the proposed system compared with other model sets.

when compared with RTL [8], 12.5% when compared with LR [9], and 5.9% when compared with HNF [26] under different use cases. The speed is improved due to the use of low-complexity temporal analysis, which assists in improving LD and non-LD identification performance under real-time scenarios. Due to these optimizations, the proposed model is able to improve the correlative analysis performance for multiple use cases and can be deployed under real-time conditions. This model also assists students in improving their performance by incrementally recommending questions as per their LD and non-LD performance levels, thereby boosting their confidence under different types of question categories.

The proposed correlation method initially collects responses from LD and non-LD students and estimates different temporal metrics. These temporal metrics are estimated on a per-category basis and correlated with new student responses. Based on this correlation, students are either upgraded to the next learning levels or given lower-level questions. Thus, this process analyzes individual students, depending on the number of attempts. It is also not necessary that there is always an improvement; there can also be a decrease in performance. The model is tracking it, which is done to inculcate continuous learning capabilities in students for different question types. This is true for both new and old students. But, specifically for new students, the model is also able to analyze:

— The ability to answer direct, indirect, vocabulary, and title questions;
— Dependency among the topics and their correlation with LD and non-LD students;
— Correctness during sequencing operations;
— Correctness to spell pseudo and sight words;
— Correctness to provide word meanings;
— Correctness to pick the odd-man-out from similar entities;
— Correctness for addition and carry propagation operations;
— Tendency to skip difficult topics;
— Too many revisits of a particular topic or set of topics;
— Time needed for individual question types.

Table 4. Correct Score of LD/Non-LD

Subject	Non-LD	LD
Basic English Grammar	7.91	4.07
Medium English Grammar	7.61	2.49
Advanced English Grammar	7.50	0.19
Comprehension: Direct	7.62	0.75
Comprehension: Inference	8.00	1.18
Comprehension: Vocabulary	6.67	1.42
Sequencing	6.34	2.29
Spelling Pseudo	6.03	1.92
Spelling Sight	6.41	3.91
Meaning	7.90	2.00
Carry Mathematics	6.21	2.24
Add Mathematics	9.15	3.81
Odd-man-out	7.75	2.2

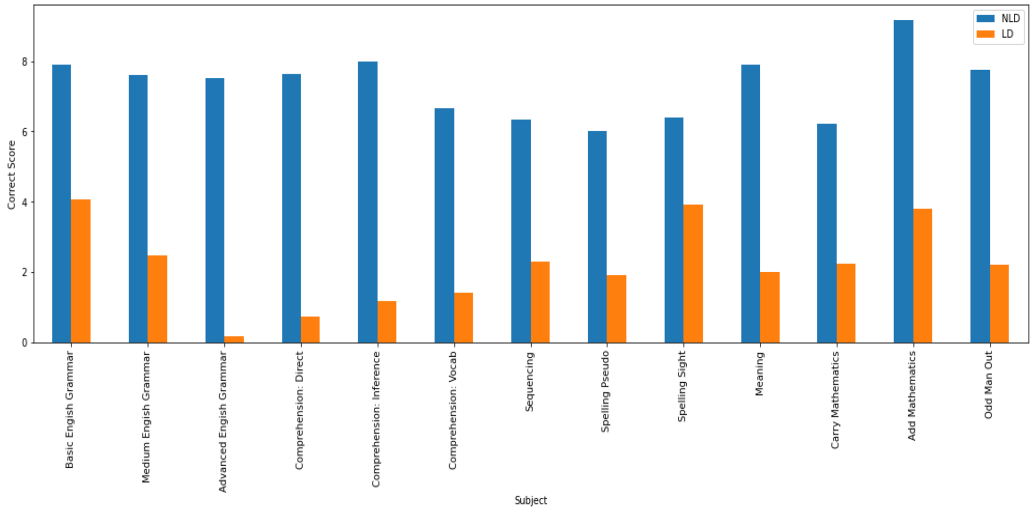


Fig. 3. Subject score of LD vs. non-LD.

As shown in Table 4 and Figure 3, the performance of LD students is 55% less than the non-LD students. It was observed that only 30% LD students attempted Medium English Grammar and only 15% LD students attempted Advanced English Grammar, inferring that the LD students are comfortable attempting only the Basic English grammar. It was also observed that the average performance of LD students in direct comprehension questions was 50% and that with inference questions was 20%. This indicates the capability of LD students to answer the question who's answers are directly found from the comprehension passage and that they find it comparatively difficult to infer the answers from the given text.

Comparing the ability to spell sight and pseudo words, it can be seen that the LD students can spell sight words with an average score of 40%, while spelling ability of the pseudo (nonsense words) is 20%. This indicates that the decoding ability of LD students is poor. Comparing the decoding capability of LD students with non-LD students, it can be seen that LD students have reduced decoding capability than that of non-LD students.

Table 5. Skips Comparisons

Subject	Non-LD	LD
Basic English Grammar	24	168
Medium English Grammar	18	145
Advanced English Grammar	12	166
Comprehension: Direct	36	52
Comprehension: Inference	21	228
Comprehension: Vocabulary	39	56
Sequencing	17	68
Spelling Pseudo	25	84
Spelling Sight	25	73
Meaning	20	84
Carry Mathematics	19	154
Add Mathematics	19	152
Odd-man-out	21	182

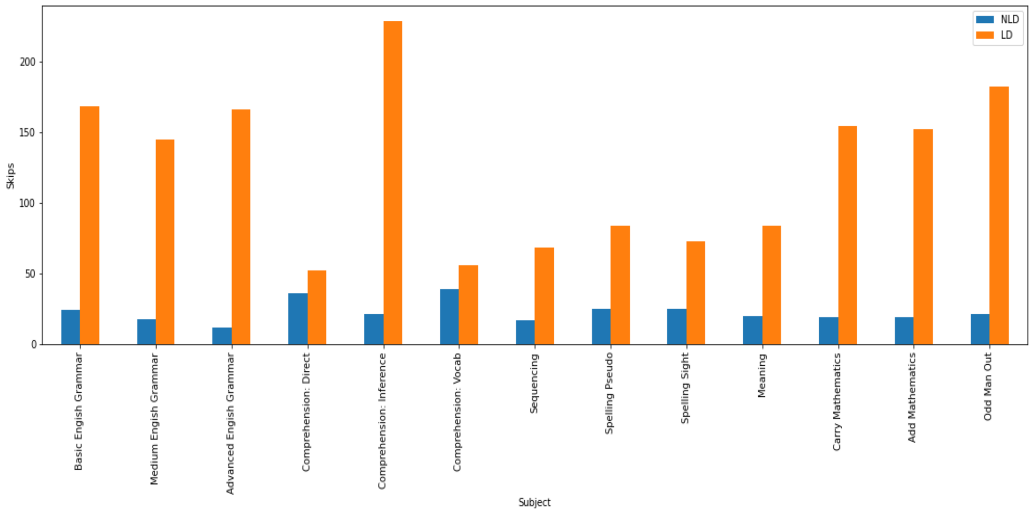


Fig. 4. Skipping tendency of LD vs. non-LD.

Skipping a particular question implies the students inability to attempt the question. Table 5 and Figure 4 show the skipping tendency of the students. It can be seen that LD students have 22% higher skipping tendency then non-LD students. It implies that the LD students need some remediation in the respective subjects so they are willing to attempt the questions instead of skipping them.

Revisiting gives students an opportunity to reflect on what they have written. This ability enables the students to correct themselves after the reflection. As seen in Table 6 and Figure 5, the revisiting tendency to a particular subject to improve his/her performance is present in non-LD students.

As can be seen from Table 7 and Figure 6, the LD students take much more time to attempt the questions than the non-LD students. LD students take more than double the time, around 130% more than that needed by non-LD students. This indicates that the LD students need remediation in this regard so they can attempt the questionnaires faster than before.

Table 6. Revisits Comparisons

Subject	Non-LD	LD
Basic English Grammar	54	0
Medium English Grammar	36	0
Advanced English Grammar	16	12
Comprehension: Direct	58	0
Comprehension: Inference	34	6
Comprehension: Vocabulary	135	6
Sequencing	0	5
Spelling Pseudo	40	1
Spelling Sight	30	6
Meaning	102	4
Carry Mathematics	199	6
Add Mathematics	42	5
Odd-man-out	170	4

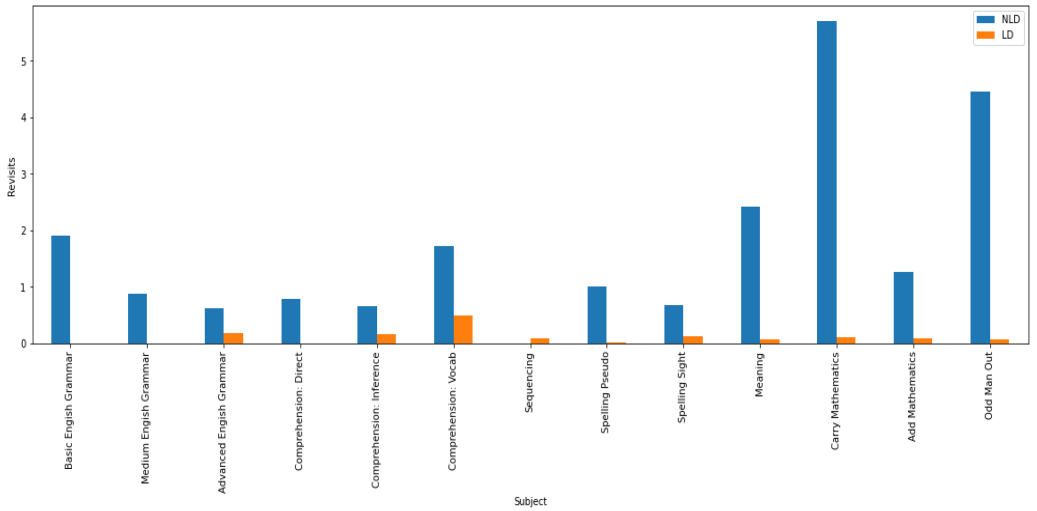


Fig. 5. Revisiting tendency of LD vs. non-LD.

The students who were facing difficulty in a particular subject were given more and adaptive questions so they perform well and their confidence is boosted. With this approach, the left graph of the figure demonstrates the count of LD/non-LD who have gained improvement, and the right side of the graph shows the same in %. Thus, it is evident that the multiple attempts of a subject in which an LD student faces difficulty will help the student in improving his skills in that subject, thereby leading to improvement. It can be seen from Tables 8, 9, and Figure 7 that the count of LD students who have achieved improvement from the provided adaptive lessons is more than that of non-LD. An average of 18 LD students have achieved an average of 33% of improvement after having multiple attempts of the adaptive lessons.

Not all students always benefit from the provided facility. From Tables 10, 11, and Figure 8, it can be seen that few students have a declined performance after multiple attempts to the same subjects. As before, the left graph in the figure shows the count of students and the right graph shows the corresponding %. Apart from direct comprehension lesson, the number of students who

Table 7. Time Comparison

Subject	Non-LD	LD
Basic English Grammar	165	476
Medium English Grammar	187	476
Advanced English Grammar	201	493
Comprehension: Direct	207	768
Comprehension: Inference	324	802
Comprehension: Vocabulary	224	738
Sequencing	482	743
Spelling Pseudo	403	656
Spelling Sight	415	748
Meaning	116	258
Carry Mathematics	449	1,146
Add Mathematics	443	845
Odd-man-out	91	460

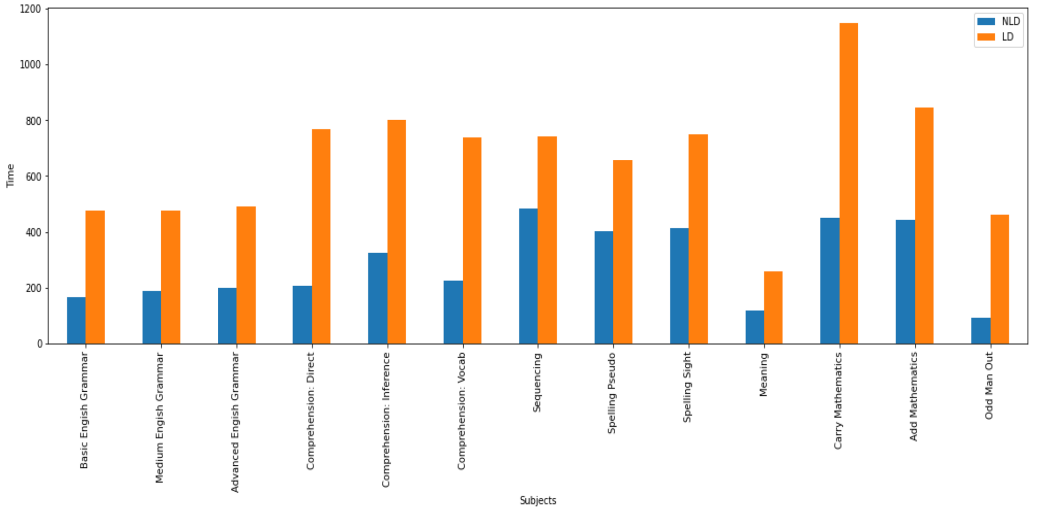


Fig. 6. Time taken by LD vs. non-LD.

have declined in performance is less than 10 and the % decline in improvement is average 5%. Due to these optimizations, the proposed model is able to improve the correlative analysis performance for multiple use cases and can be deployed under real-time conditions. This model also assists students to improve their performance via incrementally recommending questions as per their LD and non-LD performance levels, thereby boosting their confidence under different types of question categories.

7.1 Key Findings of This Research

Our research's key conclusions have important implications for educational technology and the learning experiences of students with learning difficulties. They are as follows:

(1) Better Learning Results: The suggested model's improved correlative analytical accuracy, precision, and specificity translate to better learning results. It gives teachers the ability to create unique learning ways for each student by adjusting educational content.

Table 8. Count of Students Having Improvement after Multiple Attempts

Subject	Non-LD	LD
Add Mathematics	20	3
Advanced English Grammar	1	1
Basic English Grammar	18	1
Carry Mathematics	18	0
Comprehension: Direct	12	0
Comprehension: Inference	23	0
Comprehension: Vocab	23	0
Meaning	21	1
Medium English Grammar	7	1
Odd-man-out	18	0
Sequencing	20	1
Spelling Pseudo	27	3
Spelling Sight	20	3

Table 9. Improvement (%) in Subjects after Multiple Attempts

Subject	Non-LD	LD
Add Mathematics	57	20
Advanced English Grammar	20	30
Basic English Grammar	32	60
Carry Mathematics	36	0
Comprehension: Direct	36	0
Comprehension: Inference	27	0
Comprehension: Vocab	38	0
Meaning	27	20
Medium English Grammar	33	60
Odd-man-out	39	0
Sequencing	26	20
Spelling Pseudo	26	17
Spelling Sight	35	23

(2) Efficiency Gains: The model's shorter computation time guarantees real-time flexibility, which makes it appropriate for use in realistic educational contexts. This effectiveness is essential for giving students timely support.
(3) Continuous Learning: Our results highlight the significance of allowing for both gains and losses in student performance. Regardless of a student's initial level of ability, this inclusive approach creates a possibilty of continuous improvement for all students.
(4) Targeted remedial measures are required for LD children given the performance gaps between LD and non-LD students that have been found. Multiple attempts and adaptive lessons can help LD children close the skills gap.
(5) Building Confidence: Students' confidence in answering different question types is increased by incremental recommendations based on LD and non-LD performance levels, which in turn creates a positive learning environment.

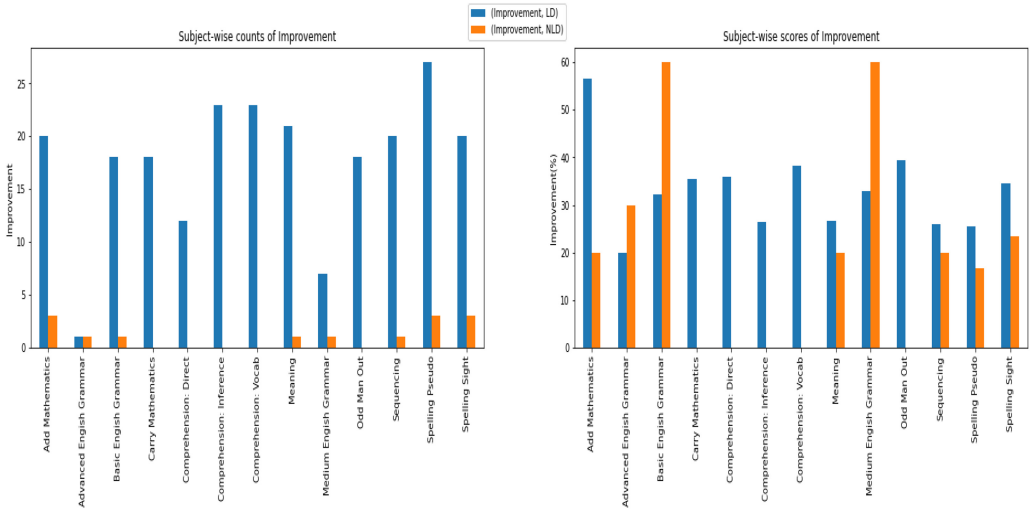


Fig. 7. Improvement in subjects after multiple attempts.

Table 10. Count of Students Having No Improvement after Multiple Attempts

Subject	Non-LD	LD
Add Mathematics	5	2
Advanced English Grammar	2	1
Basic English Grammar	6	1
Carry Mathematics	8	2
Comprehension: Direct	15	3
Comprehension: Inference	5	3
Comprehension: Vocab	4	3
Meaning	4	3
Medium English Grammar	3	0
Odd-man-out	6	3
Sequencing	6	4
Spelling Pseudo	3	3
Spelling Sight	5	2

Our suggested paradigm has a lot of scope for practical application. It can be included into educational platforms and adaptive learning systems, giving teachers insightful information and allowing them to give students the support they need. Real-time adaptation also guarantees that children get timely interventions when they require them. The model has potential for further validation and improvement on larger datasets as well as for integration with cutting-edge techniques to improve the educational experience of students in various learning environments.

8 LIMITATIONS

Although the results and contributions of our suggested model are encouraging, it is necessary to recognize a number of restrictions:

Table 11. No Improvement (%) in Subjects after Multiple Attempts

Subject	Non-LD	LD
Add Mathematics	−4	−30
Advanced English Grammar	0.00	0.00
Basic English Grammar	−3.33	0.00
Carry Mathematics	−6.25	0.00
Comprehension: Direct	−8.00	−6.67
Comprehension: Inference	0.00	−16.67
Comprehension: Vocab	0.00	−23.33
Meaning	−22.50	−36.67
Medium English Grammar	0.00	0.00
Odd-man-out	−3.33	−13.33
Sequencing	−1.67	−17.50
Spelling Pseudo	−16.67	−43.33
Spelling Sight	−4.00	−35.00

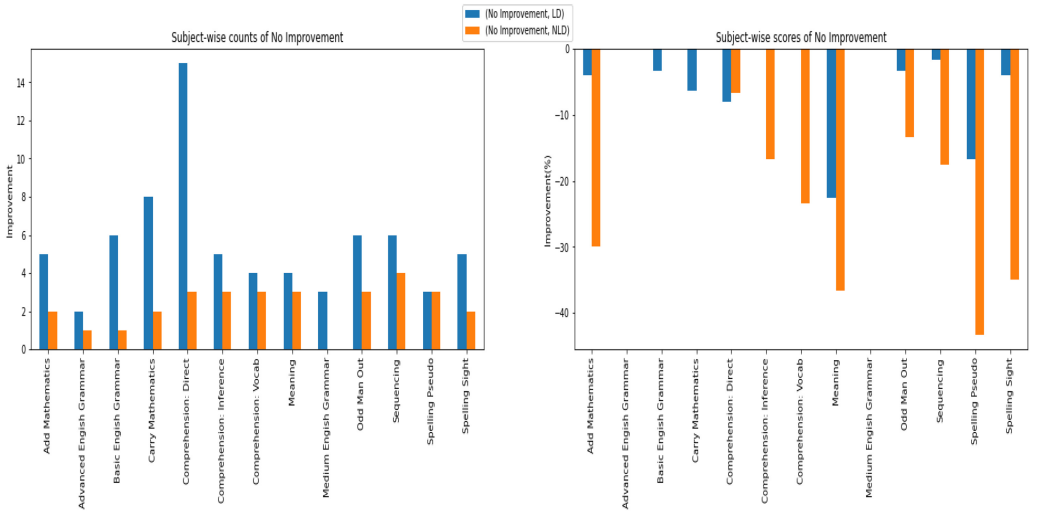


Fig. 8. No improvement in subjects after multiple attempts.

(1) Data Amount and Diversity: The amount and diversity of the data samples for our investigation may be constrained. To establish generalizability, future research should examine the model's performance on larger and more varied datasets.
(2) Model Complexity: Although the suggested model is effective, it might not be able to handle really complicated circumstances. For such situations, further optimization and scaling considerations may be required.
(3) Generalization: The model's performance has been assessed across a range of use cases, but it is important to confirm in larger contexts that the model can be applied to all educational environments.
(4) Real-world Deployment: Despite the model's promise for real-time deployment, real-world educational environment problems and factors such as hardware limitations and data privacy should be taken into account.

(5) Long-term Effects: The study mainly concentrates on immediate enhancements in learning performance. Future studies should examine how long-term benefits and sustainability of these improvements last.

Despite these drawbacks, our suggested model represents a significant advancement in the field of instructional technology and will help students with learning difficulties learn more effectively. These results can be used as a foundation for future research to improve and enhance the model's capabilities.

9 CONCLUSION AND FUTURE SCOPE

The proposed correlation method initially collects responses from students with LD and students without LD and estimates various temporal metrics. These temporal metrics are estimated per category and correlated with responses from new students. Based on this correlation, students are either promoted to higher learning levels or given questions of a lower difficulty. Consequently, this procedure analyzes each student according to their number of attempts. Also not required is an improvement; a decline in performance is also possible. The model is tracking it to instill in students the capacity for continuous learning for various question types. This is true for both new and returning students. In terms of accuracy levels, it was observed that the proposed model can improve the accuracy of correlative analysis by 14.5% when compared to RTL [8], 19.4% when compared to LR [9], and 9.0% when compared to HNF [26] for various use cases. This accuracy is enhanced as a result of the utilization of high-performance evaluation metrics, which contribute to the enhancement of LD and non-LD identification performance in real-time scenarios. In terms of consistency levels, it was observed that the proposed model improves the precision of correlative analysis by 5.9% when compared to RTL [8], 8.3% when compared to LR [9], and 3.5% when compared to HNF [26] across a variety of use cases. Correlative mappings, which aid in enhancing LD and non-LD identification performance in real-time scenarios, enhance this precision. When evaluated in terms of specificity levels, it was found that the proposed model improves recall of correlative analysis by 5.5% when compared to RTL [8], 6.4% when compared to LR [9], and 2.9% when compared to HNF [26] for various use cases. Utilizing high-performance temporal correlative mapping helps improve LD and non-LD identification performance in real-time scenarios, thereby enhancing recall. In terms of computational delay, it was found that the proposed model can improve the speed of correlative analysis by 19.4% when compared to RTL [8], 12.5% when compared to LR [9], and 5.9% when compared to HNF [26] for various use cases.

With respect to multiple attempts and correlative assessment, the performance of LD students is 55% less than the non-LD students. The skipping tendency of LD students is 22% higher than non-LD students. The number of revisits for non-LD is 220, which is far greater than mere 25 for LD students. LD students take more than double the time, around 130% more than that needed by non-LD students. This indicates that the LD students need remediation in this regard so they can attempt the questionnaires faster than before. The multiple attempts for a subject in which an LD student faces difficulty will help the student in improving his skills in that subject, thereby leading to improvement. Finally, the percentage of LD students who have achieved improvement from the provided adaptive lessons is more than that of non-LD. An average of 73% LD students has achieved an average of 33% of improvement after having multiple attempts of the adaptive lessons.

Utilizing low complexity temporal analysis aids in enhancing LD and non-LD identification performance in real-time scenarios, thereby enhancing the speed. Due to these enhancements, the proposed model can improve the performance of correlative analysis for multiple use cases and can be deployed in real-time. This model also helps students improve their performance by

incrementally recommending questions based on their LD and non-LD performance levels, thereby boosting their confidence in various question types.

Important conclusions from our study that have a big impact on educational technology and how learning for kids with learning difficulties goes are as follows:

(1) Better Learning Results: The suggested model's improved correlative analytical accuracy, precision, and specificity translate to better learning results. It gives teachers the ability to personalize learning routes for each student by adjusting educational content.
(2) Efficiency Gains: The model's shorter computation time guarantees real-time flexibility, which makes it appropriate for use in realistic educational contexts. This effectiveness is essential for giving pupils timely support.
(3) Continuous Learning: Our results highlight the significance of allowing for both gains and losses in student performance. Regardless of a student's starting ability level, this inclusive approach promotes a culture of ongoing learning for all students.
(4) Remediation for Students with LD: The observed performance gaps between LD and non-LD children highlight the necessity of individualized remedial plans. Multiple tries and adaptive lessons can help LD kids close the skills gap.
(5) Building Confidence: Students' confidence in answering different question types is increased by incremental recommendations based on LD and non-LD performance levels, which ultimately fosters a positive learning environment.

Our suggested paradigm has a lot of scope for implementation in real-world settings. It can be included into educational platforms and adaptive learning systems, giving teachers insightful information and allowing them to give students the support they need. Real-time adaptation also guarantees that pupils get timely interventions when they require them. The model has potential for further validation and improvement on larger datasets as well as for integration with cutting-edge techniques to improve the educational experience of students in various learning environments.

In the future, the performance of this model must be validated on larger data samples, and it can be enhanced through the integration of Q-Learning and Auto-encoders, which will contribute to the incremental improvement of classification performance under real-time conditions. In addition, this performance can be enhanced through the incorporation of hybrid bio-inspired models, which facilitate context-based evaluation for clinical use cases.

REFERENCES

[1] M. Modak, O. Warade, G. Saiprasad, and S. Shekhar. 2020. Machine learning based learning disability detection using LMS. In *IEEE 5th International Conference on Computing Communication and Automation (ICCCA'20)*. 414–419. DOI: 10.1109/ICCCA49541.2020.9250761

[2] E. Brito and G. P. Dias. 2020. LMS accessibility for students with disabilities: The experts' opinions. In *15th Iberian Conference on Information Systems and Technologies (CISTI'20)*. 1–5. DOI: 10.23919/CISTI49556.2020.9141046

[3] D. A. Dewantoro, M. H. M. Yasin, and M. Irvan. 2020. Measurement of e-learning readiness for students with and without disabilities. In *6th International Conference on Education and Technology (ICET'20)*. 152–159. DOI: 10.1109/ICET51153.2020.9276624

[4] A. El Kah, I. Zeroual, and A. Lakhouaja. 2021. An experimental study on the benefit of assistive technology for students with learning disabilities. In *International Conference on Digital Age and Technological Advances for Sustainable Development (ICDATA'21)*. 102–108. DOI: 10.1109/ICDATA52997.2021.00029

[5] C. Hewapathirana, K. Abeysinghe, P. Maheshani, P. Liyanage, J. Krishara, and S. Thelijjagoda. 2021. A mobile-based screening and refinement system to identify the risk of dyscalculia and dysgraphia learning disabilities in primary school students. In *10th International Conference on Information and Automation for Sustainability (ICIAfS'21)*. 287–292. DOI: 10.1109/ICIAfS52090.2021.9605998

[6] S. Tatale, N. Bhinid, R. Parmar, and S. Pcnvar. 2019. A review on virtual reality for educating students with learning disabilities. In *5th International Conference on Computing, Communication, Control and Automation (ICCUBEA'19)*. 1–6. DOI: 10.1109/ICCUBEA47591.2019.9128570

[7] S. Tatale, N. Bhirud, R. Parmar, and S. Pawar. 2019. Education using virtual reality for students with learning disabilities. In *5th International Conference on Computing, Communication, Control and Automation (ICCUBEA'19)*. 1–7. DOI: 10.1109/ICCUBEA47591.2019.9128515

[8] D. B. Hajovsky, K. E. Maki, S. R. Chesnut, C. A. Barrett, and M. K. Burns. 2022. Specific learning disability identification in an RtI Method: Do measures of cognitive ability matter? *Learn. Disab. Res. Pract.* 37, 4 (2022), 280–293. DOI: https://doi.org/10.1111/ldrp.12292

[9] K. Chen, M. Huang, X. Zhu, and G. Wang. 2021. Learning disability early warning system based on classification algorithm. In *2nd International Conference on Information Science and Education (ICISE-IE'21)*. 597–600. DOI: 10.1109/ICISE-IE53922.2021.00141

[10] S. Burgstahler. 2021. Designing online learning to be accessible to students with disabilities. In *Conference on Research in Equitable and Sustained Participation in Engineering, Computing, and Technology (RESPECT'21)* 1–2. DOI: 10.1109/RESPECT51740.2021.9620647

[11] Mohammad A. Alzubaidi, Mwaffaq Otoom, and Nouran S. Ahmad. 2021. Real-time assistive reader pen for Arabic language. *ACM Trans. Asian Low-resour. Lang. Inf. Process.* 20, 1 (2021), 1–30.

[12] S. Wang, Z. Wang, W. Che, S. Zhao, and T. Liu. 2021. Combining self-supervised learning and active learning for disfluency detection. *Trans. Asian Low-resour. Lang. Inf. Process.* 21, 3 (2021), 1–25.

[13] R. L. Taylor and L. Sternberg. 1989. Teaching students with learning disabilities. In *Exceptional Children*. Springer, New York, NY. DOI: https://doi.org/10.1007/978-1-4612-3602-3_4

[14] C. Nilvius and I. Svensson. 2021. Efficacy evaluation of a full-scale response to intervention program for enhancing student reading abilities in a Swedish school context. *Read Writ* 35 (2022), 1239–1264. DOI: https://doi.org/10.1007/s11145-021-10237-3

[15] J. Seale, C. Colwell, T. Coughlan, T. Heiman, D. Kaspi-Tsahor, and Olenik-Shemesh. 2021. "Dreaming in colour": Disabled higher education students' perspectives on improving design practices that would enable them to benefit from their use of technologies. *Educ. Inf. Technol.* 26, 1687–1719. DOI: https://doi.org/10.1007/s10639-020-10329-7

[16] L. Didion, J. R. Toste, and S. A. Benz. 2020. Self-determination to increase oral reading fluency performance: Pilot and replication single-case design studies. *Learn. Disab. Res. Pract.* 35 (2020), 218–231. DOI: https://doi.org/10.1111/ldrp.12234

[17] J. Dietrichson, T. Filges, R. H. Klokker, B. C. A. Viinholt, M. Bøg, and U. H. Jensen. 2020. Targeted school-based interventions for improving reading and mathematics for students with, or at risk of, academic difficulties in grades 7–12: A systematic review. *Campbell System. Rev.* 17, 2 (2021), e1152. DOI: https://doi.org/10.1002/cl2.1152

[18] K. González-Arias, J. Gálvez-Quezada, V. Robles-Bykbaev, F. Bermeo-Zambrano, and A. León-Pesántez. 2021. An intelligent decision support system for educational inclusion of children with disabilities: An approximation for multidisciplinary teams. In *IEEE World Conference on Engineering Education (EDUNINE'21)*. 1–6. DOI: 10.1109/EDUNINE51952.2021.9429110

[19] J. B. Schumaker. 2022. Lessons learned during the development and validation of an intensive evidence-based reading intervention for secondary students. *Learn. Disab. Res. Pract.* 37, 4 (2022), 294–313. DOI: https://doi.org/10.1111/ldrp.12293

[20] D. Fuchs, D. Mock, P. L. Morgan, and C. L. Young. 2003. Responsiveness-to-intervention: Definitions, evidence, and implications for the learning disabilities construct. *Learn. Disab. Res. Pract.* 18 (2003), 157–171. DOI: https://doi.org/10.1111/1540-5826.00072

[21] P.-G. Jung, K. L. McMaster, A. K. Kunkel, J. Shin, and P. M. Stecker. 2018. Effects of data-based individualization for students with intensive learning needs: A meta-analysis. *Learn. Disab. Res. Pract.* 33, 144–155. DOI: https://doi.org/10.1111/ldrp.12172

[22] J. Agrawal and L. L. Morin. 2016. Evidence-based practices: Applications of concrete representational abstract framework across math concepts for students with mathematics disabilities. *Learn. Disab. Res. Pract.* 31 (2016), 34–44. DOI: https://doi.org/10.1111/ldrp.12093

[23] W. J. Therrien, J. C. Taylor, J. L. Hosp, E. R. Kaldenberg, and J. Gorsh. 2011. Science instruction for students with learning disabilities: A meta-analysis. *Learn. Disab. Res. Pract.* 26 (2011), 188–203. DOI: https://doi.org/10.1111/j.1540-5826.2011.00340.x

[24] S. P. Miller and P. J. Hudson. 2007. Using evidence-based practices to build mathematics competence related to conceptual, procedural, and declarative knowledge. *Learn. Disab. Res. Pract.* 22 (2007), 47–57. DOI: https://doi.org/10.1111/j.1540-5826.2007.00230.x

[25] P. McCardle, H. S. Scarborough, and H. W. Catts. 2001. Predicting, explaining, and preventing children's reading difficulties. *Learn. Disab. Res. Pract.* 16 (2001), 230–239. DOI: https://doi.org/10.1111/0938-8982.00023

[26] T. Lackaye, M. Margalit, O. Ziv, and T. Ziman. 2006. Comparisons of self-efficacy, mood, effort, and hope between students with learning disabilities and their non-LD-matched peers. *Learn. Disab. Res. Pract.* 21 (2006), 111–121. DOI: https://doi.org/10.1111/j.1540-5826.2006.00211.x

[27] Yi Li. 2022. Analysis of psychological disorders and adaptive influence of blended learning of college students. *Computat. Intell. Neurosci.* 2022, Article ID 5418738 (2022), 12 pages. DOI: https://doi.org/10.1155/2022/5418738

[28] Lin Wending. 2022. Implementing the hybrid neuro-fuzzy system to model specific learning disability in special university education programs. *J. Math.* 2022, Article ID 6540542 (2022), 7 pages. DOI: https://doi.org/10.1155/2022/6540542

[29] Kanchana Menikdiwela. 2020. Student misbehavior: An exploratory study based on Sri Lankan secondary school teachers' perceptions. *J. Educ. Pract.* 11 (2020), 103–113.

[30] V. Mahlangu. 2022. Exploring the supervision of gifted students in open distance e-learning setting in higher education context: University of South Africa. *J. Gift. Educ. Creativ.* 9, 1 (2022), 57–74. Retrieved from https://dergipark.org.tr/en/pub/jgedc/issue/68496/1062986

[31] M. S. A. B. Rosli, I. S. Isa, S. A. Ramlan, S. N. Sulaiman, and M. I. F. Maruzuki. 2021. Development of CNN transfer learning for dyslexia handwriting recognition. In *11th IEEE International Conference on Control System, Computing and Engineering (ICCSCE'21)*. 194–199. DOI: 10.1109/ICCSCE52189.2021. 9530971

[32] I. Sarah, K. Soundarya, R. Dhanalakshmi, and T. Deenadayalan. 2020. DYS-I-CAN: An aid for the dyslexic to improve the skills using mobile application. In *International Conference on System, Computation, Automation and Networking (ICSCAN'20)*. 1–5. DOI: 10.1109/ICSCAN49426.2020.9262375

[33] R. Kariyawasam, M. Nadeeshani, T. Hamid, I. Subasinghe, and P. Ratnayake. 2019. A gamified approach for screening and intervention of dyslexia, dysgraphia and dyscalculia. In *International Conference on Advancements in Computing (ICAC'19)*. 156–161. DOI: 10.1109/ICAC49085.2019.9103336

[34] N. Giri, T. Saini, K. Bhole, A. Bhosale, T. Shetty, A. Subramanyam, and S. Shelke. 2020. Detection of dyscalculia using machine learning. In *5th International Conference on Communication and Electronics Systems (ICCES'20)*. 1–6. DOI: 10.1109/ICCES48766.2020.9137871

[35] R. M. Abdou, N. H. Hamouda, and A. M. Fawzy. 2020. Validity and reliability of the Arabic dyscalculia test in diagnosing Egyptian dyscalculic school-age children. *Egypt J. Otolaryngol.* 36, 18 (2020). DOI: https://doi.org/10.1186/s43163-020-00020-6

[36] S. Bugden, L. Peters, N. Nosworthy, L. Archibald, and D. Ansari. 2021. Identifying children with persistent developmental dyscalculia from a 2-min test of symbolic and nonsymbolic numerical magnitude processing. *Mind, Brain, Educ.* 15 (2021), 88–102. DOI: https://doi.org/10.1111/mbe.12268

Received 6 May 2023; revised 29 October 2023; accepted 1 November 2023